CStats - Lab #5

March 24, 2026

Exercise 1.
In order to test whether four brands of gasoline give equal performance in terms of mileage, each of three cars was driven with each of the four brands of gasoline. Then each of the $(3)(4) = 12$ possible combinations was repeated four times. The number of miles per gallon for each of the four repetitions in each cell is recorded in Table 1. Test the hypotheses H_{AB}: no interaction, H_A: no row effect, and H_B: no column effect, each at the 5% significance level.

	Brand 1	**Brand 2**	**Brand 3**	**Brand 4**
Car 1	31.0, 24.9, 26.2, 28.8	26.3, 30.0, 25.2, 31.6	25.8, 29.4, 24.5, 24.8	27.8, 27.3, 28.2, 30.4
Car 2	30.6, 29.5, 30.8, 28.9	25.5, 26.8, 27.4, 29.4	26.6, 23.7, 28.2, 26.1	28.1, 27.1, 31.5, 29.1
Car 3	24.2, 23.1, 26.8, 27.4	27.4, 28.1, 26.4, 26.9	25.2, 26.7, 27.7, 28.1	26.3, 26.4, 27.9, 28.8

Table 1: Gasoline Performance by Car and Brand

Solution 1.
To analyze this data, we need to conduct a two-way analysis of variance (ANOVA) to test already given three hypotheses in the problem:

- H_{AB}: No interaction between car type and gasoline brand.
- H_A: No row effect (no difference in mileage across different car types)
- H_B: No column effect (no difference in mileage across different gasoline brands)

Then, we should define a null hypothesis and alternative hypothesis for each situation.

- Interaction effect H_{AB}
  - H_0: There is no interaction between car type and gasoline brand (mileage differences are additive).

- H_1: There is an interaction effect, meaning the effect of gasoline brand on mileage depends on the car type.

- Row effect H_A
  - H_0: The mean mileage does not differ significantly among the three cars.
  - H_1: At least one car has significantly different mileage than the others.

- Column Effect H_B
  - H_0: The mean mileage does not differ significantly among the four gasoline brands.
  - H_1: At least one gasoline brand has significantly different mileage than the others.

Since we have two categorical factors, which are Car Type and Gasoline Brand, we use a two-way ANOVA which is written by

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

where y_{ijk} is mileage for the kth repetition of car i and gasoline j, μ is overall mean mileage, α_i is effect of car i which is row effect, β_j is effect of gasoline brand j which is column effect, $(\alpha\beta)_{ij}$ interaction effect between car i and gasoline brand j and ϵ_{ijk} is random error.

A two-way ANOVA produces an ANOVA table with the following key values:

Two-way ANOVA Table

Source	SS	DF	MS	F	p
A (Car Effect)	SS_A	$r-1$	$MS_A = SS_A/df_A$	F_A	p_A
B (Brand Effect)	SS_B	$c-1$	$MS_B = SS_B/df_B$	F_B	p_B
Interaction(H_{AB})	SS_{AB}	$(r-1)(c-1)$	$MS_{AB} = SS_{AB}/df_{AB}$	F_{AB}	p_{AB}
Error	SSE	$rc(n-1)$	MSE		
Total	SST	$N-1$			

where $r = 3$ is number of car types, $c = 4$ is number of gasoline brands, $n = 4$ is the number of repetitions per cell and $N = 48$ is number of observations. Here are decision rules:

- If $p_{AB} < 0.05$, reject H_{AB} (there is an interaction effect)
- If $p_A < 0.05$, reject H_A (there is a significant difference in mileage among cars)
- If $p_B < 0.05$, reject H_B (there is a significant difference in mileage among gasoline brands)

Let calculate the values in the two-way ANOVA table. At first, let define the terms as the following:

$$r = 3 \quad \text{(Number of Cars)}$$
$$c = 4 \quad \text{(Number of Gasoline Brands)}$$
$$n = 4 \quad \text{(Number of Repetitions per Cell)}$$
$$N = r \times c \times n = 3 \times 4 \times 4 = 48 \quad \text{(Total Number of Observations)}$$

Let:

- y_{ijk} = mileage for car i, brand j, and replication k.
- $\bar{y}$ = Grand Mean (overall mean of all values).
- $\bar{y}_i$= Row Mean (mean mileage for car i).
- $\bar{y}_j$ = Column Mean (mean mileage for brand j).
- $\bar{y}_{ij}$ = Cell Mean (mean mileage for car i and brand j).

$$\bar{Y} = \frac{1}{N} \sum_{i=1}^{r} \sum_{j=1}^{c} \sum_{k=1}^{n} y_{ijk} = \frac{\text{sum of all observations}}{N} = \frac{1318.9}{48} = 27.4770$$

Total Sum of Squares:

$$SS_{\text{Total}} = \sum_{i=1}^{r} \sum_{j=1}^{c} \sum_{k=1}^{n} (y_{ijk} - \bar{y})^2 = 198.8448 \tag{1}$$

To calculate SS_A, we need the mean of each car.

- Mean Car 1: 27.6375
- Mean Car 2: 28.08125
- Mean Car 3: 26.7125

Then, Sum of Squares for Car Effect SS_A is:

$$SS_A = c \times n \sum_{i=1}^{r} (\bar{y}_i - \bar{y})^2 = 15.6054. \tag{2}$$

Similarly, here are means of brands:

- Mean Brand 1: 27.6833
- Mean Brand 2: 27.5833

- Mean Brand 3: 26.4000
- Mean Brand 4: 28.2417

Sum of Squares for Brand Effect SS_B is written by

$$SS_B = r \times n \sum_{j=1}^{c} (\bar{y}_{.j} - \bar{y})^2 = 21.5823. \tag{3}$$

Here is to calculate $\bar{y}_{ij}$, which is cell means:

- Mean Car 1 - Brand 1: 27.7250
- Mean Car 1 - Brand 2: 28.2750
- Mean Car 1 - Brand 3: 26.1250
- Mean Car 1 - Brand 4: 28.4250
- Mean Car 2 - Brand 1: 29.9500
- Mean Car 2 - Brand 2: 27.2750
- Mean Car 2 - Brand 3: 26.1500
- Mean Car 2 - Brand 4: 28.9500
- Mean Car 3 - Brand 1: 25.3750
- Mean Car 3 - Brand 2: 27.2000
- Mean Car 3 - Brand 3: 26.9250
- Mean Car 3 - Brand 4: 27.3500

Then, Sum of Squares for Interaction SS_{AB} is:

$$SS_{AB} = n \sum_{i=1}^{r} \sum_{j=1}^{c} (\bar{y}_{ij} - \bar{y}_i - \bar{y}_j + \bar{y})^2 = 36.1246 \tag{4}$$

Sum of Squares for Error SSE is:

$$SSE = SST - SS_A - SS_B - SS_{AB} = 125.5325 \tag{5}$$

Degrees of freedoms are as the followings:

$$
\begin{aligned}
df_A &= r - 1 = 2 \\
df_B &= c - 1 = 3 \\
df_{AB} &= (r-1)(c-1) = 6 \\
df_E &= r \times c \times (n-1) = 36 \\
df_T &= N - 1 = 47
\end{aligned}
$$

Mean Squares are:

$$
\begin{aligned}
MS_A &= \frac{SS_A}{df_A} = \frac{15.6054}{2} = 7.8027 \\
MS_B &= \frac{SS_B}{df_B} = \frac{21.5823}{3} = 7.1941 \\
MS_{AB} &= \frac{SS_{AB}}{df_{AB}} = \frac{36.1246}{6} = 6.0208 \\
MSE &= \frac{SSE}{df_E} = \frac{125.5325}{36} = 3.4870
\end{aligned}
$$

F-statistics are as

$$
\begin{aligned}
F_A &= \frac{MS_A}{MSE} = 2.2376 \\
F_B &= \frac{MS_B}{MSE} = 2.0631 \\
F_{AB} &= \frac{MS_{AB}}{MSE} = 1.7266
\end{aligned}
$$

Finally, p-values are:

$$
\begin{aligned}
p_A &= 1 - F_{(F_A, df_A, df_E)} = 0.1213 \\
p_B &= 1 - F_{(F_B, df_B, df_E)} = 0.1224 \\
p_{AB} &= 1 - F_{(F_{AB}, df_{AB}, df_E)} = 0.1430
\end{aligned}
$$

Then, we obtain the filled two-way ANOVA table as

Two-way ANOVA Table

Source	SS	DF	MS	F	p
A	15.6054	2	7.8027	2.2376	0.1213
B	21.5823	3	7.1941	2.0631	0.1224
Interaction	36.1246	6	6.0208	1.7266	0.1430
Error	125.5325	36	3.4870		
Total	198.8448	47			

Therefore, since $p = 0.1430$ which is greater than 0.05, there is no significant interaction between car type and gasoline brand (fails to reject H_{AB}). We accept H_A which means no significant difference in mileage across different cars because $p = 0.1213$ which is greater than 0.05. In addition, $p = 0.1224$ then accept H_B, which means no significant difference in mileage across different gasoline brands.

Exercise 2.
The output in Table 2 was obtained from a computer program that performed a two-factor ANOVA on a factorial experiment.
(a) Fill in the blanks in the ANOVA table. You can use bounds on the p-values.
(b) How many levels were used for factor B?
(c) How many replicates of the experiment were performed?
(d) What conclusions would you draw about this experiment?

Two-way ANOVA: y **versus** A, B

Source	DF	SS	MS	F	p
A	1	?	0.0002	?	?
B	?	180.378	?	?	?
Interaction	3	8.479	?	?	0.932
Error	8	158.797	?		
Total	15	347.653			

Table 2: Two-way ANOVA Table

Solution 2.
(a) Here are the given information:

- Degrees of Freedom for Total $df_T = 15$
- Degrees of Freedom for Error $df_E = 8$
- Degrees of Freedom for Interaction $df_{AB} = 3$
- Sum of Squares for $SS_B = 180.378$.
- Sum of Squares for Interaction $SS_{AB} = 8.479$.
- Sum of Squares for Error $SSE = 158.797$.
- The p-value for the Interaction $p_{AB} = 0.932$.
- Total Sum of Squares $SST = 347.653$.

We know that:

$$\mathrm{df}_T = \mathrm{DF}_A + \mathrm{df}_B + \mathrm{df}_{AB} + \mathrm{df}_E.$$

Then $\mathrm{df}_T = 1 + (\mathrm{df}_B) + 3 + 8 = 15$. So, $\mathrm{df}_B = 3$.

We write the relationship between the sum of squares as the following and put the known values into it. When we solve it for SS_A, we get

$\mathrm{SST} = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SS}_{AB} + \mathrm{SSE}$

$347.653 = \mathrm{SS}_A + 180.378 + 8.479 + 158.797 \quad \mathrm{SS}_A = 347.653 - 180.378 - 8.479 - 158.797 = 0.000$

To calculate MS_A,

$$MS_A = \frac{\mathrm{SS}_A}{\mathrm{df}_A} = \frac{0}{1} = 0.$$

Similarly,

$$MS_B = \frac{\mathrm{SS}_B}{\mathrm{df}_B} = \frac{180.378}{3} = 60.126.$$

Then, for MS_{AB}, we write the following:

$$MS_{AB} = \frac{\mathrm{SS}_{AB}}{\mathrm{df}_{AB}} = \frac{8.479}{3} = 2.826.$$

Also,

$$MSE = \frac{SSE}{\mathrm{df}_E} = \frac{158.797}{8} = 19.8496.$$

F-statistics for A and AB are computed by:

$$F_A = \frac{MS_A}{MSE} = \frac{0}{19.8496} = 0.$$

$$F_{AB} = \frac{MS_{AB}}{MSE} = \frac{2.826}{19.8496} = 0.1427.$$

Since the F-statistic for A is 0 (indicating no effect of A), the p-value is very large and close to 1. The p-value for $F_B = 3.03$ corresponds to a p-value of around 0.06 which is greater than 0.05. Given that $F_{AB} = 0.1427$, the p-value for the interaction is approximately 0.932.

Therefore, completed two-way ANOVA table as the following:

Two-way ANOVA Table

Source	SS	DF	MS	F	p
A	1	0	0.0002	0	1.000
B	3	180.378	60.126	3.03	0.060
Interaction	3	8.479	2.826	0.1427	0.932
Error	8	158.797	19.8496		
Total	15	347.653			

(b) Factor B has 4 levels, as $df_B = 3$ and the number of levels is $df_B + 1 = 4$.
(c) There are 2 replicates of the experiment, as df_E is obtained by multiplying $n-1$ by total number of factor combinations, where $n = 2$.
(d)

- Factor A does not appear to have a significant effect (p-value = 1.000).
- Factor B has a marginally significant effect on the response variable (p-value is approximately 0.06).
- There is no significant interaction between factors A and B (p-value = 0.932).

Exercise 3.
Consider the two-factor model as

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

for $i = 1, 2, ..., a$, $j = 1, 2, ..., b$ and $k = 1, 2, ..., r$ where r is the number of replications. Assuming all the factors are fixed, write down the analysis of variance table. If we have only one replicate, what would you use as the "experimental error" to test hypotheses?

Solution 3.
Here is the three-way ANOVA table:

Two-way ANOVA Table

Source	SS	DF	MS	F	p
A	SS_A	$a-1$	$MS_A = SS_A/df_A$	F_A	p_A
B	SS_B	$b-1$	$MS_B = SS_B/df_B$	F_B	p_B
Interaction	SS_{AB}	$(a-1)(b-1)$	$MS_{AB} = SS_{AB}/df_{AB}$	F_{AB}	p_{AB}
Error	SSE	$ab(r-1)$	MSE		
Total	SST	$abr-1$			

To test the hypotheses for Factor A, Factor B, and their interaction, we use the mean square error (MSE) as the "experimental error." The corresponding F-statistics are computed as follows:

$$F_A = \frac{MS_A}{MSE}, \quad \text{to test the main effect of Factor A,} \tag{6}$$

$$F_B = \frac{MS_B}{MSE}, \quad \text{to test the main effect of Factor B,} \tag{7}$$

$$F_{AB} = \frac{MS_{AB}}{MSE}, \quad \text{to test the interaction effect.} \tag{8}$$

If there is only one replicate ($r = 1$), then the error degrees of freedom become:

$$ab(r-1) = ab(1-1) = 0.$$

Since there are no replications, there is no independent estimate of the experimental error (MSE). In this case, the interaction term (MS_{AB}) is often used as the error term if the interaction is assumed negligible. If interaction effects are significant, then no valid test for the main effects can be conducted using ANOVA.

Exercise 4.
The experiment shown in Table 3 is a variation of the study on polysilicon doping. The response variable is base current. Is there evidence (with 0.05) indicating that either polysilicon doping level or anneal temperature affects base current? Finally, define a model for this experiment and conclude your results.

Polysilicon Doping (ions)	Anneal Temperature (°C)		
	900	950	1000
1×10^{20}	4.60	10.15	11.01
	4.40	10.20	10.58
2×10^{20}	3.20	9.38	10.81
	3.50	10.02	10.60

Table 3: Data - Polysilicon doping vs Anneal Temperature

Solution 4.
Let start by modeling the base current y using two factors such that x_1 is Polysilicon doping level and x_2 is Anneal temperature. We treat this as a two-factor experiment with an interaction term. We perform ANOVA to determine whether the factors x_1 and x_2 and their interaction significantly influence the base current.

The given data table provides the base current values for different combinations of doping level and anneal temperature. Then we can reorganize it as the following:

Polysilicon Doping (ions)	Anneal Temperature (°C)	Base Current (y)
1×10^{20}	900	$4.60, 4.40$
1×10^{20}	950	$10.15, 10.20$
1×10^{20}	1000	$11.01, 10.58$
2×10^{20}	900	$3.20, 3.50$
2×10^{20}	950	$9.38, 10.02$
2×10^{20}	1000	$10.81, 10.60$

We need to calculate the means for each group: Let start with Mean for doping level 1×10^{20}.

$$\text{For Anneal Temp 900:} \quad \frac{4.60 + 4.40}{2} = 4.50$$
$$\text{For Anneal Temp 950:} \quad \frac{10.15 + 10.20}{2} = 10.175$$
$$\text{For Anneal Temp 1000:} \quad \frac{11.01 + 10.58}{2} = 10.795$$

Similarly, Mean for doping level 2×10^{20} is written by the following:

$$\text{For Anneal Temp 900:} \quad \frac{3.20 + 3.50}{2} = 3.35$$
$$\text{For Anneal Temp 950:} \quad \frac{9.38 + 10.02}{2} = 9.70$$
$$\text{For Anneal Temp 1000:} \quad \frac{10.81 + 10.60}{2} = 10.705$$

Also, $\bar{y} = 8.20$ which is overall mean. To assess whether the factors x_1 (doping level) and x_2 (anneal temperature) affect the base current, we conduct an ANOVA test. The null hypothesis for each factor is that there is no significant effect (i.e., $\beta_1 = 0$ and $\beta_2 = 0$). The sum of squares are computed as follows:

$$SS_A = n \sum_i (\bar{y_i} - \bar{y})^2 = 0.98$$
$$SS_B = m \sum_j (\bar{y_j} - \bar{y})^2 = 111.19$$
$$SS_{AB} = n \sum_{i,j} (\bar{y_{ij}} - \bar{y_i} - \bar{y_j} + \bar{y})^2 = 0.58$$
$$SST = \sum_{i,j} (y_{ij} - \bar{y})^2 = 113.13$$
$$SS_E = SST - (SS_A + SS_B + SS_{AB}) = 0.39$$

where $\bar{y_i}$ is the mean of the i-th level of doping, and n is the number of observations per doping group, $\bar{y_j}$ is the mean of the j-th level of temperature, $\bar{y_{ij}}$ is the mean for the combination of x_1 and x_2.

SS_A represents the variation due to the polysilicon doping level (x_1). SS_B represents the variation due to the anneal temperature. SS_{AB} represents the interaction effect between doping level and anneal temperature. SSE represents the residual variation after accounting for the factors and their interaction.

Here are the degrees of freedoms:

- $df_T = N - 1 = 12 - 1 = 11$, where N is the total number of observations.
- $df_A = a - 1 = 2 - 1 = 1$, where a is the number of levels of factor A (polysilicon doping).
- $df_B = b - 1 = 3 - 1 = 2$, where b is the number of levels of factor B (anneal temperature).
- $df_{AB} = (a - 1)(b - 1) = 2$, for the interaction term.
- $df_E = df_T - (df_A + df_B + df_{AB}) = 6$.

Mean squares are as the followings:

- $MS_A = \frac{SS_A}{df_A} = 0.98$
- $MS_B = \frac{SS_B}{df_B} = 55.59$
- $MS_{AB} = \frac{SS_{AB}}{df_{AB}} = 0.29$
- $MSE = \frac{SSE}{df_E} = 0.06$

Then, F-statistics are:

$$F_A = \frac{MS_A}{MSE} = 15.26, \quad F_B = \frac{MS_B}{MSE} = 865.16, \quad F_{AB} = \frac{MS_{AB}}{MSE} = 4.48$$

Therefore, p-values are the following:

- $p_A = 0.0079$
- $p_B < 0.0001$
- $p_{AB} = 0.0645$

Let calculate the values for the model of this test which is given by

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \epsilon$$

where β_0 is intercept, β_1 is affect of doping level (x_1), β_2 is effect of anneal temperature (x_2), β_{12} is interaction effect of doping level and anneal temperature and ϵ is random error from normal distribution. The followings are the values for the model:

- $SS_{Model} = SST - SSE = 112.74$
- $df_{Model} = df_A + df_B + df_{AB} = 5$
- $MS_{Model} = \frac{SS_{Model}}{df_{Model}} = 22.55$
- $F_{Model} = \frac{MS_{Model}}{MSE} = 350.91$
- $p_{Model} = 1 - F_{(F_{Model}, df_{Model}, df_E)} < 0.0001$

The model F-value of 350.91 implies the model is significant. Since $p < 0.05$ for the model, it means that the model terms is significant, that is, there is a significant effect between polysilicon doping and anneal temperature.